Circuits

Building Models to Describe our World

Modeling and Solving Circuits, and Kirchoff's Laws

Outline

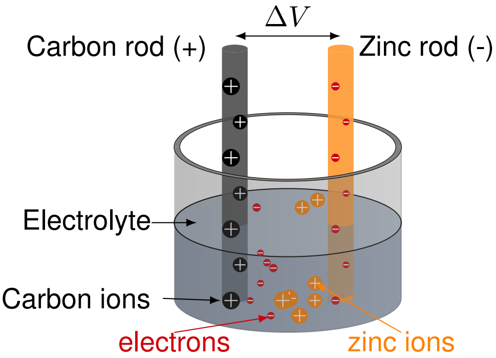
Modeling Circuits

Building Models to Describe our World

Modeling and Solving Circuits, and Kirchoff's Laws

The electric cell

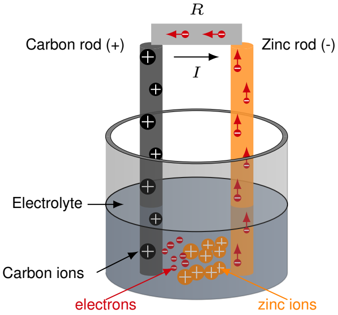
- Zinc “likes” to be dissolved as an ion in acid.
- The zinc ions leave their electrons behind in the electrode, making the zinc electrode negatively charged.
- Electrons that are loosely bound in a carbon electrode are attracted into the electrolyte solution, leaving the carbon electrode positively charged.
- Very quickly, the process stops as an equilibrium is reached, with a net electric potential difference between electrodes.



An electric cell made with an electrolyte, such as H_2SO_4 (sulfuric acid) and two metals.

Current flow in a battery

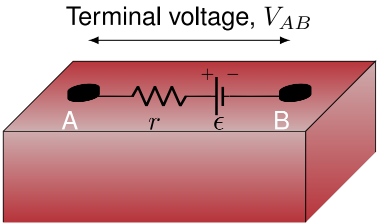
- The rate of chemical reactions in the battery limits the current that can be produced.
- The components of the battery themselves will have a finite resistance (e.g. the rods).
- If the resistance of the resistor (and battery) is low enough, the rate of chemical reactions cannot sustain the current implied by Ohm's Law ($V = RI$). The voltage of the battery thus drops until the current given by Ohm's Law can be sustained by the chemical reactions.



As electrons leave the zinc, more ions can dissolve. They eventually neutralize in the solution as electrons come in through the carbon.

Modeling a real battery

- We can model a real battery as an ideal battery + a resistor (“internal resistance”).
- The “emf”, ϵ, is the potential difference of the ideal battery:
  - It is a constant, independent of what we connect to the battery.
  - “Internal voltage” of the battery, if you prefer a better description than “electromotive force”.
- The “terminal voltage” is the actual voltage measured at the external terminals of the battery. It *depends on the current flowing through the battery* (which results in a voltage drop across the internal resistance, r).

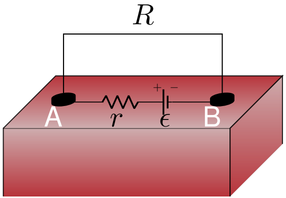


A real battery, modelled with an internal resistance.

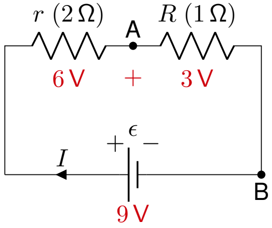
Terminal voltage in a real circuit

- The current through both resistors is $3\,\text{A}$.
- The voltage drop across a resistor, $V = RI$:
  - The voltage drop across the $2\,\Omega$ resistor is $(2\,\Omega)(3\,\text{A}) = 6\,\text{V}$
  - The voltage drop across the $1\,\Omega$ resistor is $(1\,\Omega)(3\,\text{A}) = 3\,\text{V}$
- We identify that the "terminal voltage" is between the $-$ terminal (B) of the ideal battery, and the point between the two resistors (A)
- The terminal voltage is $V = 9\,\text{V} - 6\,\text{V} = 3\,\text{V}$ *assuming the internal battery can sustain* $3\,\text{A}$.

With a larger resistor, R, the total current would be smaller and the voltage drop across r negligible.



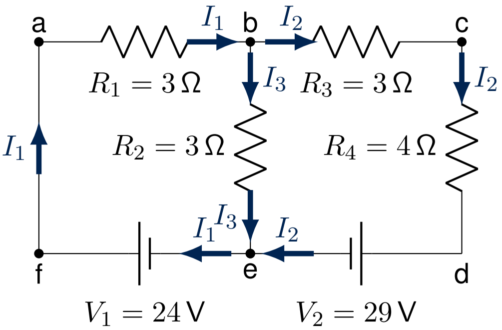
A battery shorted with a wire of resistance, R



Circuit diagram for the shorted battery. The current is $I = \frac{9\,\text{V}}{(2\,\Omega)+(4\,\Omega)} = 3\,\text{A}$

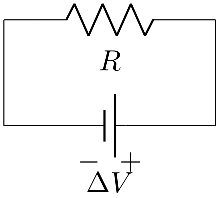
Analysing circuits

- If you know the currents through all of the resistors in a circuit, you know everything that there is to know about the circuit:
  - Given the currents in all resistors, you can calculate the voltage across each of the resistors.
- Usually, the easiest method to do this is to combine resistors together into equivalent resistors.
- However, this cannot always be done, in particular, when the circuit contains more than one battery.



We would like to know all of the currents in this circuit. It cannot be simplified to a single equivalent resistor.

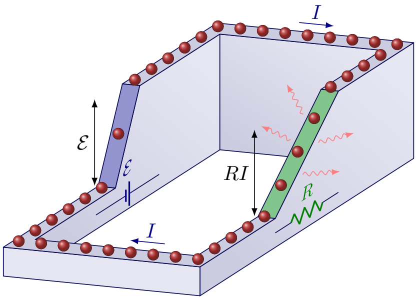
A not so great analogy



- Often, the analogy between a circuit and a roller coaster is made:
  - The battery acts like the lift for the train, giving it potential energy.
  - The train then looses its potential energy by going down the track, which is like the resistor.
- Why is this not a great analogy?

A better analogy

- If the charges are like the cars on a roller coaster, the battery is the lift that brings them up the hill.
- When they go down the hill, the key difference with a roller coaster is that they arrive at the bottom with zero speed (they lost all of their energy in the resistor $\rightarrow$ like a paddle wheel).



Charges “lose” their energy as heat in the resistors.

A better analogy

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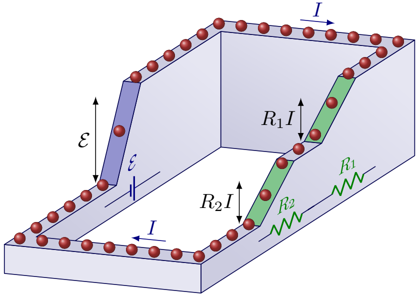

With resistors in series, the same current goes through each resistor, and the potential energy is shared between the two.

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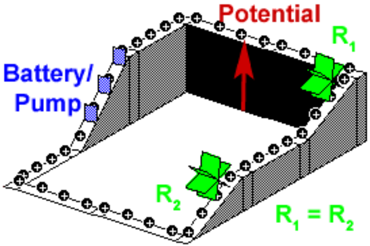

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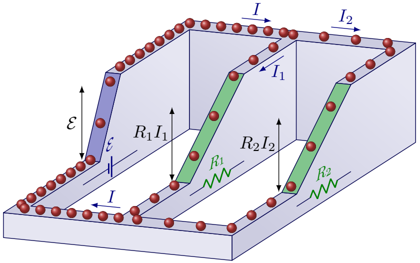

With resistors in parallel, there is less current through each resistor but the same amount of potential energy.

A better analogy

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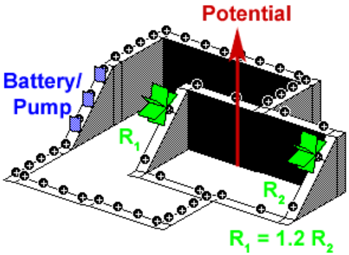
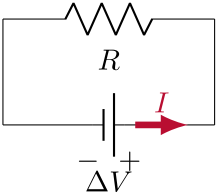


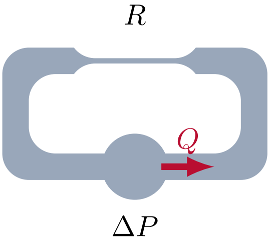
Or even better

- Change in pressure drives the water in a pipe.
- The resistance to water flow in the pipe depends on how long and how narrow the pipe is:
  - Longer $\rightarrow$ more resistance.
  - Wider $\rightarrow$ less resistance.
- Just like a resistor:

$$R\rho\frac{L}{A}$$



Current is driven through the resistor by a potential difference. The current is the same everywhere in the wire (continuity).



Fluid is driven through the resistor by a difference in pressure (e.g. from a pump). The flow rate is the same everywhere in the pipe (continuity).

Kirchhoff's rules

- **Charge conservation:**
  - Charges cannot just appear or disappear from a circuit, they have to go somewhere (there is nowhere for water to enter or leave a closed system).
- **Energy conservation:**
  - If a charge moves around a loop in a circuit, its energy must not change (the water does not flow faster and faster in a closed system).

A garden fountain is a closed system. There are multiple paths for the water to go down (resistors in parallel). Since there is no paddle wheel, what is the water's kinetic energy being converted to?

Kirchoff’s Laws

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Kirchhoff's rules

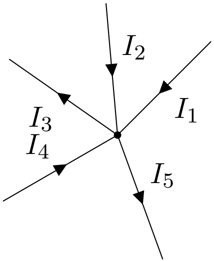
- **Junction rule** (charge):
  - The sum of the currents coming into a junction must equal the sum of the currents coming out of a junction:

    incoming current $=$ outgoing current

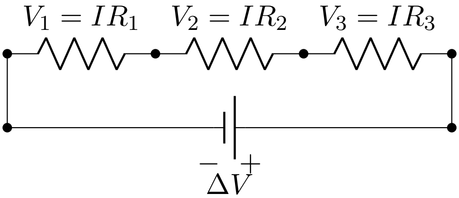
    $$I_1 + I_2 + I_4 = I_3 + I_5$$

- **Loop rule** (energy):
  - The sum of the potential differences in any closed loop must be zero:

    $$V_1 + V_2 + V_3 + \Delta V = 0$$



The junction rule.

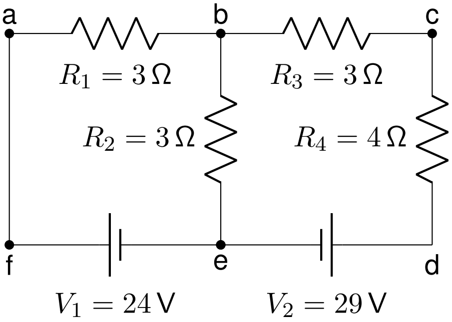


The loop rule.

Using Kirchhoff's Laws

Getting started

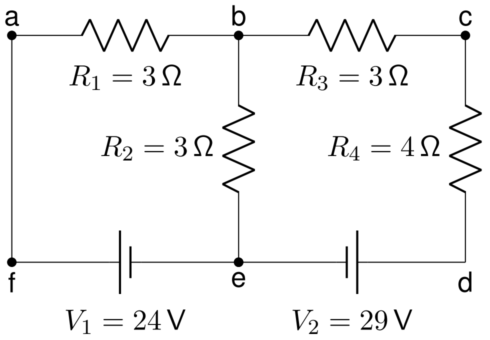
1. Make a well-labeled diagram.
2. Identify the junctions.
3. Identify the loops.



We would like to know all of the currents in this circuit.

Using Kirchhoff's Laws

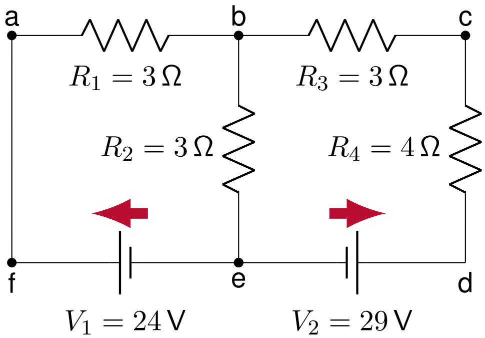
1. Draw a good diagram with nodes labeled.



We want to know all current in this circuit diagram.

Using Kirchhoff's Laws

1. Draw a good diagram with nodes labeled.
2. Draw "battery arrows" from $-$ to $+$.



We want to know all current in this circuit diagram.

Using Kirchhoff's Laws

1. Draw a good diagram with nodes labeled.
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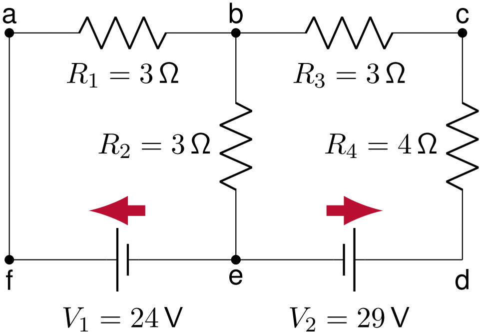
3. Make a **consistent** *guess* for currents.



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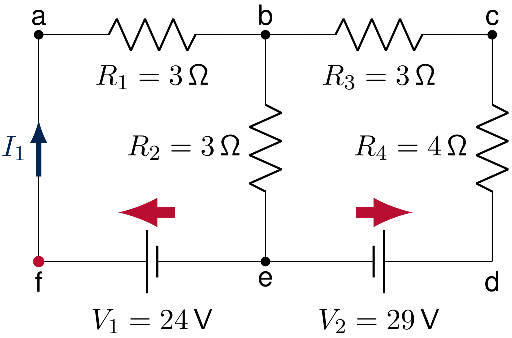
Using Kirchhoff's Laws

1. Draw a good diagram with nodes labeled.
2. Draw "battery arrows" from $-$ to $+$.
3. Make a **consistent** *guess* for currents.
   :
   - Start *anywhere* and keep introducing a new current each time you hit a junction.



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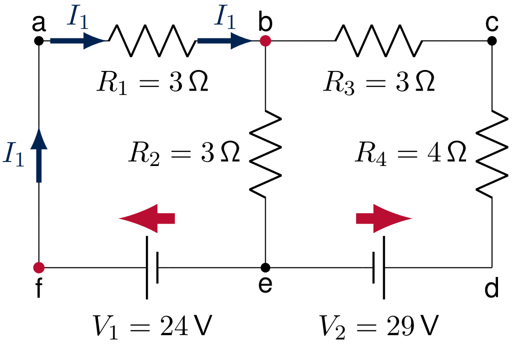

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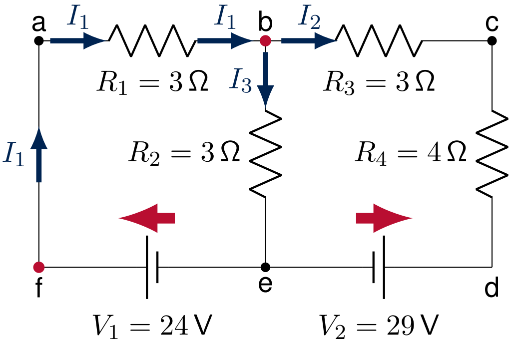

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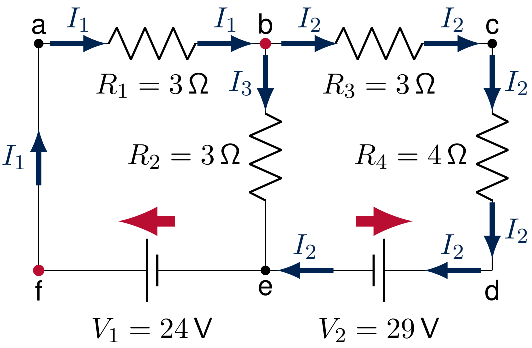

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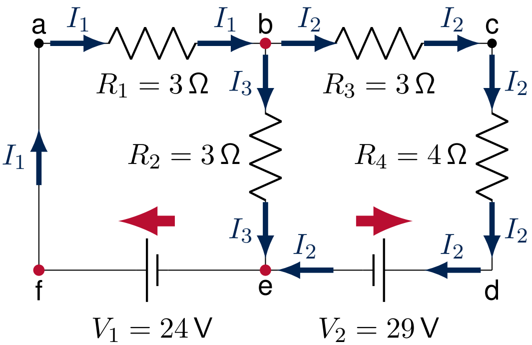

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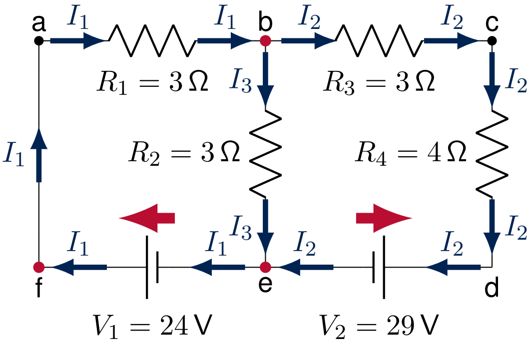


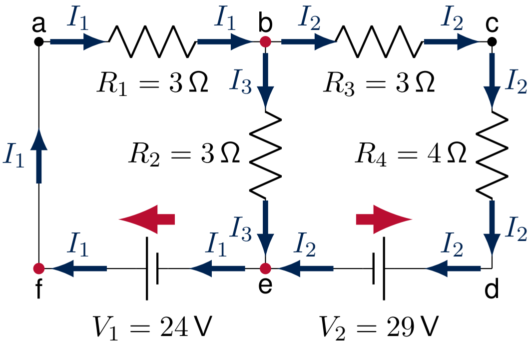
We want to know all current in this circuit diagram.

Using Kirchhoff's Laws

1. Draw a good diagram with nodes labeled.
2. Draw "battery arrows" from $-$ to $+$.
3. Make a **consistent** *guess* for currents.
4. Write the Junction Rule (once per junction, up to $N-1$ for N junctions):

$$\boxed{I_1 = I_2 + I_3}$$

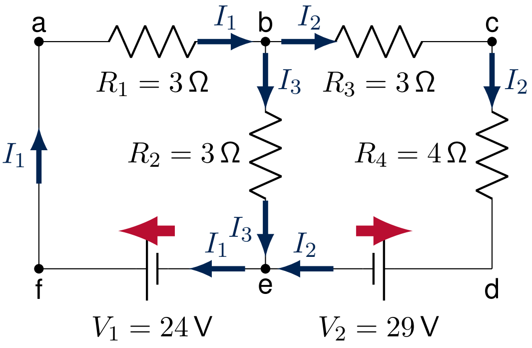
Expect 3 unknowns, so need 2 more equations... The loop equations!



We want to know all current in this circuit diagram.

The loop equations

- **Important**: *You choose* which way to "walk" around the loop. It's completely arbitrary. Two rules:
    1. When you encounter a **battery arrow: Add the voltage** if you walk in the same direction as the arrow, subtract otherwise.
    2. When you encounter a **resistor: Subtract the voltage** across the resistor if you walk in the same direction as the current, add otherwise
- Across a closed loop, the voltage must be zero.



Circuit diagram with everything well-labeled.

Solving Circuits

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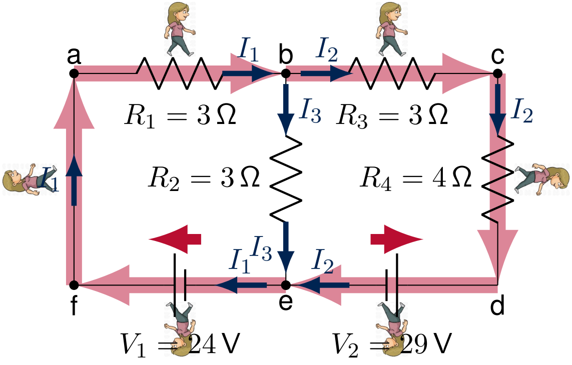
The loop equations

→ Consider loop $abcdefa$:

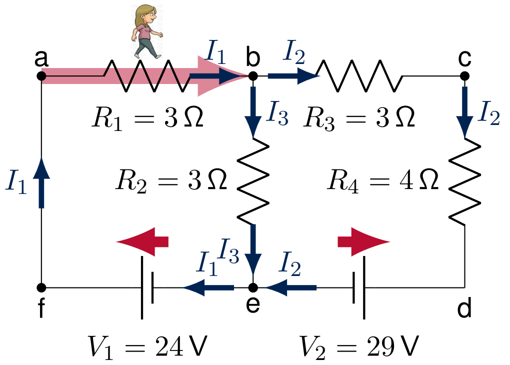


Going around loop $(abcdefa)$.

The loop equations

→ Consider loop $abcdefa$:

- (ab): with current, I_1 through resistor R_1: $-R_1 I_1 \to -3I_1$



Going around loop $(abcdefa)$.

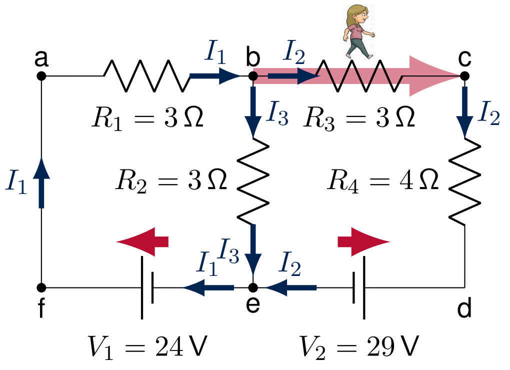
Equation for loop $(abcdefa)$:

$$-3I_1 \ldots$$

The loop equations

$\rightarrow$ Consider loop $abcdefa$:

- (bc): with current, I_2 through resistor R_3: $-R_3 I_2 \rightarrow -3I_2$



Going around loop $(abcdefa)$.

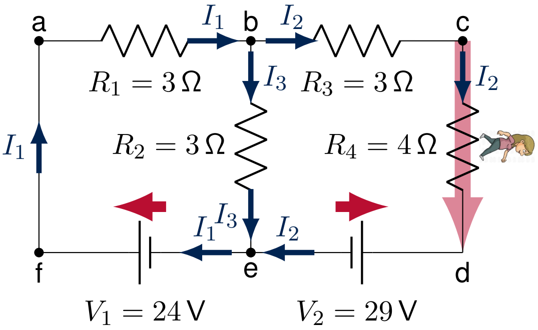
Equation for loop $(abcdefa)$:

$$-3I_1 \; -3I_2 \ldots$$

The loop equations

→ Consider loop $abcdefa$:

- (cd): with current, I_2 through resistor R_4: $-R_3 I_2 \rightarrow -4I_2$



Going around loop $(abcdefa)$.

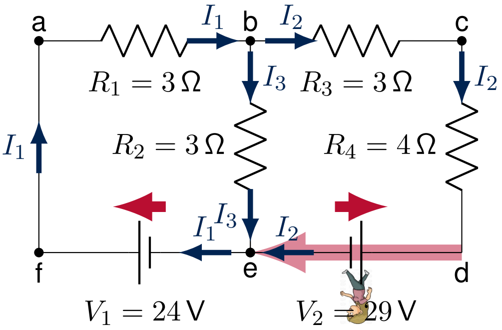
Equation for loop $(abcdefa)$:

$$-3I_1 - 3I_2 - 4I_2 \ldots$$

The loop equations

→ Consider loop $abcdefa$:

- (de): *against* battery arrow:
  $-V_2 \to -29$



Going around loop $(abcdefa)$.

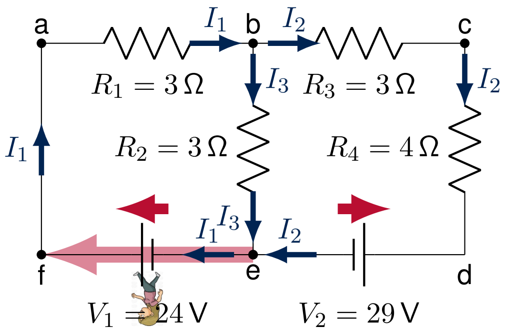
Equation for loop $(abcdefa)$:

$$-3I_1 - 3I_2 - 4I_2 - 29 \ldots$$

The loop equations

→ Consider loop $abcdefa$:

- (ef): with battery arrow: $+V_1 \to +24$



Going around loop $(abcdefa)$.

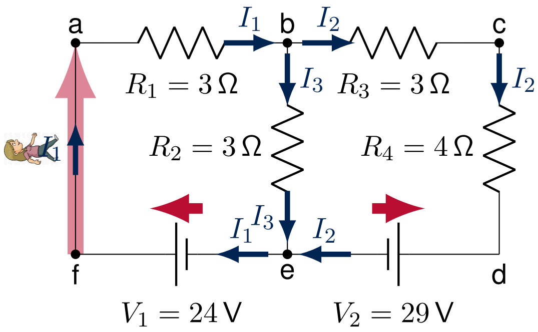
Equation for loop $(abcdefa)$:

$$-3I_1 - 3I_2 - 4I_2 - 29 + 24 \ldots$$

The loop equations

→ Consider loop $abcdefa$:

- (fa): no battery or resistor, so no voltage change. Back to starting point: $= 0$



Going around loop $(abcdefa)$.

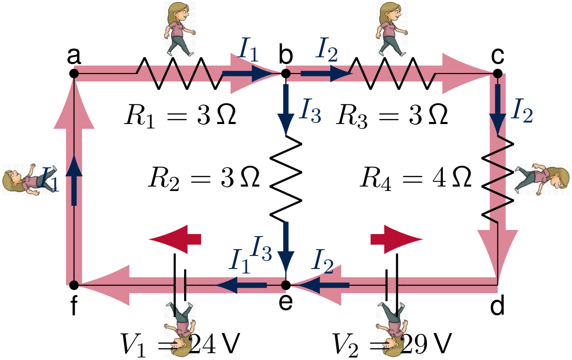
Equation for loop $(abcdefa)$:

$$-3I_1 - 3I_2 - 4I_2 - 29 + 24 = 0$$

The loop equations

→ Consider loop $abcdefa$:

- We now have a second equation (recall that there are 3 unknowns). We need one more loop equation.



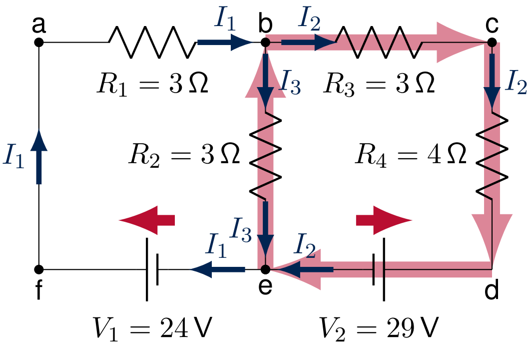
Going around loop $(abcdefa)$.

Equation for loop $(abcdefa)$:

$$\therefore -3I_1 - 3I_2 - 4I_2 - 29 + 24 = 0$$

The loop equations

- Choose loop $bcdeb$:
  - (bc): $-R_3 I_2 \rightarrow -3I_2$ (with current through resistor)
  - (cd): $-R_4 I_2 \rightarrow -4I_2$ (with current through resistor)
  - (de): $-V_2 \rightarrow -29$ (*against* battery arrow)
  - (eb): $+R_2 I_3 \rightarrow +3I_3$ (*against* current through resistor)



Circuit diagram with loop $(bcdeb)$ highlighted.

Equation for loop $(bcdeb)$:

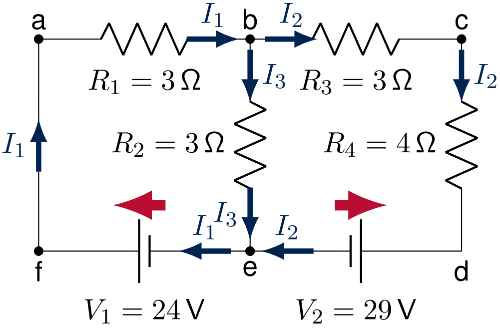
$$\therefore -3I_2 - 4I_2 - 29 + 3I_3 = 0$$

Solving the circuit

- We now have 3 equations and 3 unknowns, allowing us to determine all of the currents.

$$
\begin{aligned}
I_2 + I_3 &= I_1 && \text{(Junction rule)} \\
-3I_1 - 3I_2 - 4I_2 - 29 + 24 &= 0 && \text{(Loop abcdefa)} \\
-3I_2 - 4I_2 - 29 + 3I_3 &= 0 && \text{(Loop bcdeb)}
\end{aligned}
$$

- The system of linear equations is easier to handle if you don't include the units (so make sure everything is in SI units!).



Well-labeled circuit diagram.

Solving a circuit - Foolproof method!

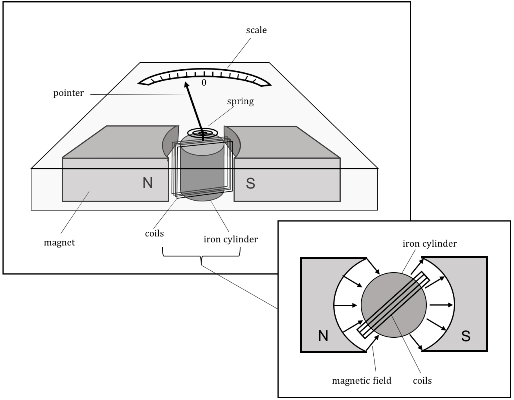
1. Make a good diagram.
2. Simplify resistors that can be combined.
3. Draw “battery arrows”, from $-$ to $+$ (small slide to big side).
4. Guess currents in a consistent manner (start somewhere and proceed).
5. Write the junction rule equations (once per junction).
6. Write the loop equations.
7. Solve N equations for N unknown currents.

Direction of current(s)

It is possible that you find negative values for the current(s). In that case, **first, solve for all of the other currents**. Then, *at the very end*, reverse the arrows in the diagram since you guessed the direction wrong (but in a consistent way!).

The galvanometer

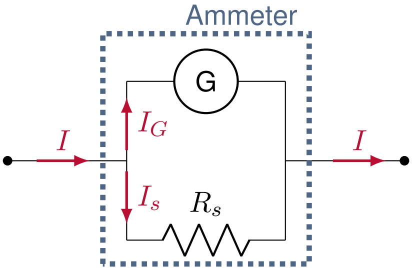
- As we will learn, a wire that carries a current will feel a force if placed in a magnetic field.
- A galvanometer is a very sensitive device for measuring current.
- Typically, they measure currents of order micro amperes.
- Can obviously use to measure current.
- Can also use to measure voltage.



A galvanometer

Measuring current

- Based on using a galvanometer:
  - $\rightarrow$ Need to have a very small current through the galvanometer
- An **ammeter** measures currents, even potentially large ones.
- We thus add a resistor in **parallel** with a galvanometer to create an ammeter.
- This is called a *shunt* resistor.



An ammeter made using a galvanometer and a (small) shunt resistor. Most of the current goes through the shunt. An ammeter is connected in series and has a very low resistance.

Measuring current

- We know the current through the galvanometer and the value of all resistors. The total current is:

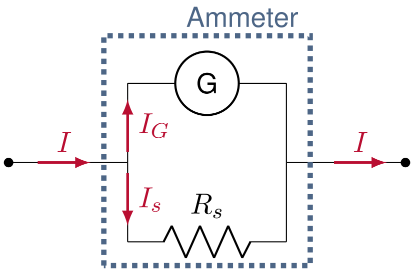
$$I = I_G + I_s$$

- Both resistors have the same potential difference:

$$R_G I_G = R_s I_s$$

- The total current, in terms of I_G, is thus:

$$I = I_G + I_s = I_G \left(1 + \frac{R_G}{R_s}\right)$$

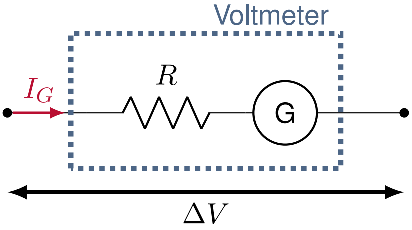


An ammeter made using a galvanometer and a (small) shunt resistor. Most of the current goes through the shunt. An ammeter is connected in series and has a very low resistance.

Measuring Voltage

- Again, we need a small current through the galvanometer, but would like to measure a large potential difference.
- We place a large resistor in **series** with the galvanometer.
- Both see the same current, so if R is large it will have the biggest potential difference.

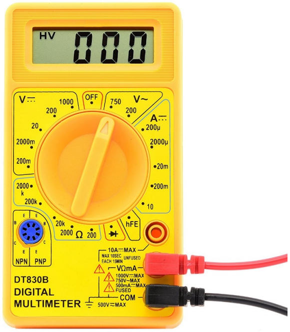
$$\Delta V = I_G(R + R_G)$$



A voltmeter made using a galvanometer and a (large) resistor in series. Most of the voltage drop happens across the large resistor, R. A voltmeter is connected in parallel and has a very large resistance.

The multimeter

- Can measure current, voltage and resistance.
- The knob chooses the value of the resistor and its configuration.
- To measure resistance, the device has an internal voltage source; it then measures the current across a resistor connected across it leads (the resistor cannot be connected to anything else).



A typical multimeter.

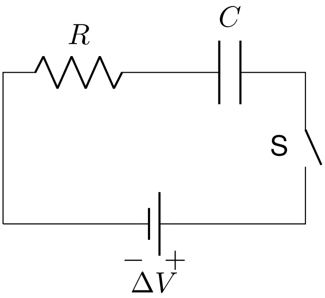
RC Circuits

- Consider a circuit with a resistor and a capacitor:
  - Current cannot flow through a capacitor.
  - Capacitor charges up with time, until current is zero.

  → The current in an RC circuit is not constant!

  Consider the loop rule (recall $Q = CV$ for a capacitor:

$$\Delta V - IR - \frac{Q}{C} = 0$$
$$\Delta V - \frac{dQ}{dt}R - \frac{Q}{C} = 0$$
$$\therefore Q(t) = C\Delta V\left(1 - e^{-\frac{t}{RC}}\right)$$



A simple circuit with a resistor, battery, and capacitor.

When the switch is closed, current decreases exponentially with time:

$$I(t) = \frac{dQ}{dt} = \frac{\Delta V}{R}e^{-\frac{t}{RC}}$$